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Batch: 15

Experiment

Question

Write a menu-driven C program to read a **variable-length square integer array** and perform the following operations using separate functions:

1. Determine whether all the elements on its main diagonal are distinct.
2. Find the sum of its main diagonal elements.
3. Display its elements using a diagonal traversal. Begin with the element in the top-left corner and traverse successive diagonals alternately in the upward-right and downward-left directions.
4. Exit.

Algorithm

1. Start.
2. Read the order n of the square matrix.
3. If $n \leq 0$, display an error message and ask again.
4. Declare a variable-length array $a[n][n]$.
5. Read all the elements of the matrix.
6. Display the menu repeatedly:
 - o Main diagonal distinct check
 - o Sum of main diagonal
 - o Diagonal traversal
 - o Exit
7. Read the user's choice.
8. Use a switch statement:
 - o **Choice 1:** Call `checkDistinct()`.
 - o **Choice 2:** Call `diagonalSum()`.
 - o **Choice 3:** Call `diagonalTraversal()`.
 - o **Choice 4:** Exit.
 - o Otherwise, display "Invalid choice".
9. In `checkDistinct()`:
 - o Compare each main diagonal element $a[i][i]$ with the diagonal elements after it.
 - o If any two are equal, the diagonal elements are not distinct.
10. In `diagonalSum()`:
 - o Add $a[i][i]$ for $i = 0$ to $n-1$.
11. In `diagonalTraversal()`:

- Consider diagonals according to `row + column`.
 - Traverse successive diagonals in alternating directions.
12. Repeat the menu until the user selects Exit.
 13. Stop.

Test Case Table

TC	Matrix	Choice	Expected Output
1	1 2 3 / 4 5 6 / 7 8 9 1		Main diagonal 1 5 9 → All distinct
2	1 2 3 / 4 5 6 / 7 8 9 2		Sum = 15
3	1 2 3 / 4 5 6 / 7 8 9 3		1 2 4 7 5 3 6 8 9
4	1 2 3 / 4 1 6 / 7 8 9 1		Main diagonal 1 1 9 → Not distinct
5	Any valid matrix	5	Invalid choice
6	5	1	Main diagonal 5 → All distinct
7	5	2	Sum = 5

C Program

```
#include <stdio.h>

void checkDistinct(int n, int a[n][n])
{
    int i, j, distinct = 1;

    printf("Main diagonal elements: ");

    for (i = 0; i < n; i++)
        printf("%d ", a[i][i]);

    for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (a[i][i] == a[j][j])
            {
                distinct = 0;
                break;
            }
        }
    }

    if (distinct == 0)
        break;
}

if (distinct == 1)
```

```

        printf("\nAll the main diagonal elements are distinct.\n");
    else
        printf("\nAll the main diagonal elements are not distinct.\n");
}

```

```

void diagonalSum(int n, int a[n][n])
{
    int i, sum = 0;

    for (i = 0; i < n; i++)
    {
        sum = sum + a[i][i];
    }

    printf("Sum of main diagonal elements: %d\n", sum);
}

```

```

void diagonalTraversal(int n, int a[n][n])
{
    int d, i, j;

    printf("Diagonal traversal: ");

    for (d = 0; d <= 2 * n - 2; d++)
    {
        if (d % 2 == 0)
        {
            /* Downward-left direction */
            if (d < n)
                j = d;
            else
                j = n - 1;

            i = d - j;

            while (i < n && j >= 0)
            {
                printf("%d ", a[i][j]);
                i++;
                j--;
            }
        }
        else
        {
            /* Upward-right direction */
            if (d < n)
                i = d;
            else

```

```

        i = n - 1;

        j = d - i;

        while (i >= 0 && j < n)
        {
            printf("%d ", a[i][j]);
            i--;
            j++;
        }
    }

    printf("\n");
}

```

```

int main()
{
    int n, i, j, choice;

    do
    {
        printf("Enter the order of square matrix: ");
        scanf("%d", &n);

        if (n <= 0)
            printf("Invalid input. Enter a positive value.\n");

    } while (n <= 0);

    int a[n][n];

    printf("Enter the matrix elements:\n");

    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    do
    {
        printf("\nMENU\n");
        printf("1. Check whether main diagonal elements are distinct\n");
        printf("2. Find sum of main diagonal elements\n");
        printf("3. Diagonal traversal\n");
        printf("4. Exit\n");
    }
}

```

```
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        checkDistinct(n, a);
        break;

    case 2:
        diagonalSum(n, a);
        break;

    case 3:
        diagonalTraversal(n, a);
        break;

    case 4:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice.\n");
}

} while (choice != 4);

return 0;
}
```